Week 6-2

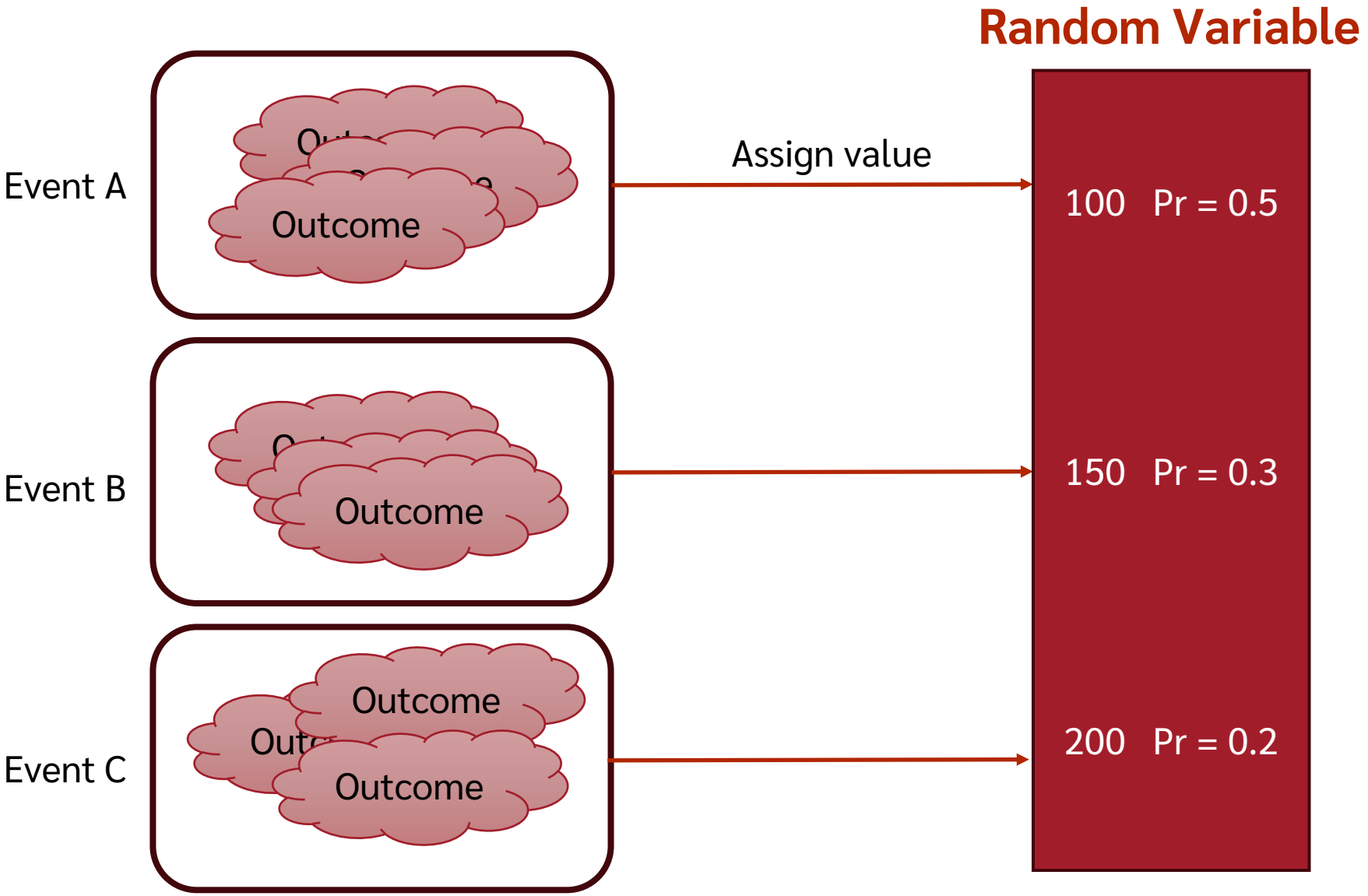
Continuous Random Variable

September 12, 2024

2110205 STATS COMP ENG

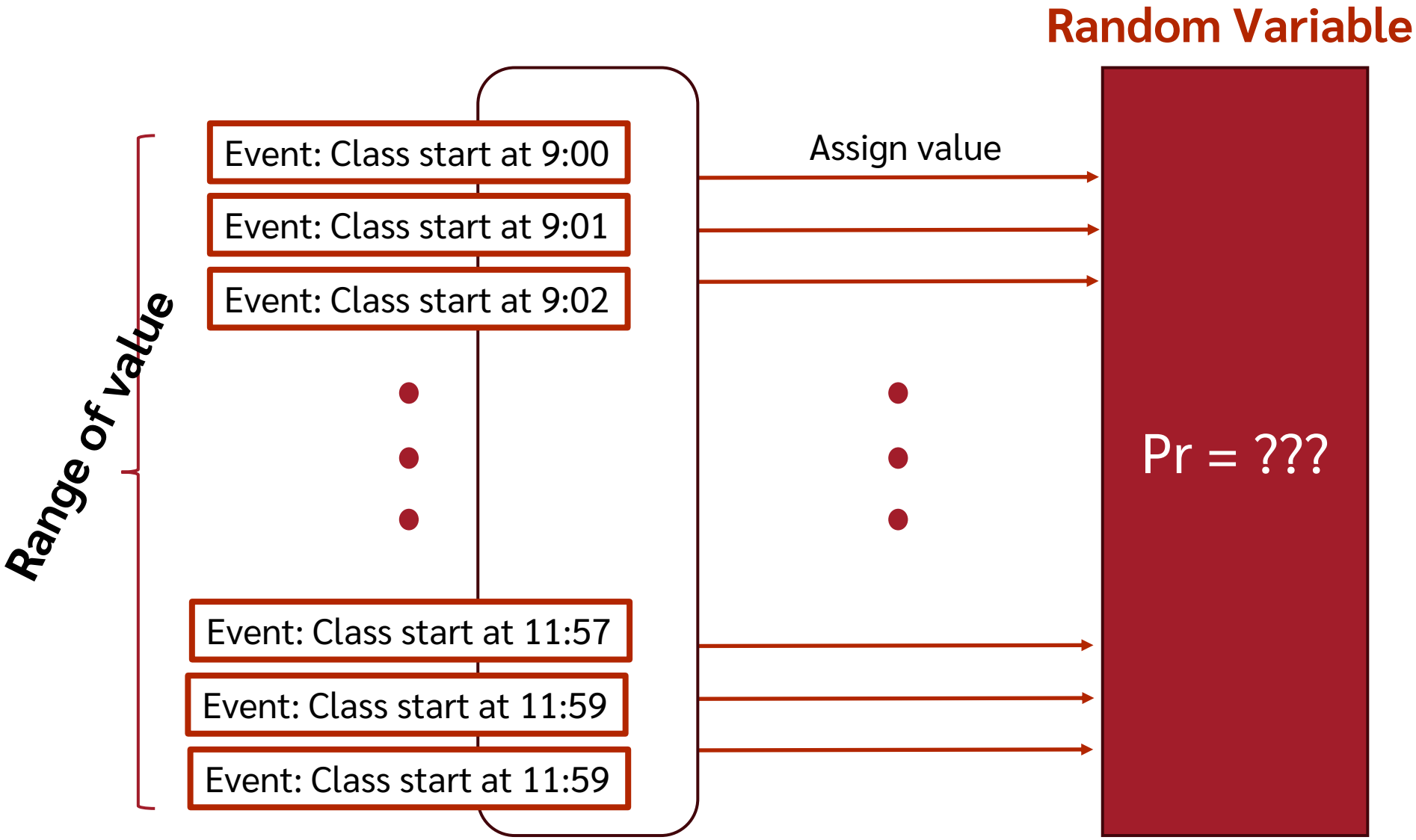
CHULA
COMPUTER
ΣNGINEERING

Discrete Random Variable



Value DEPEND on the outcome.

Continuous Random Variable



Value DEPEND on the outcome.

Examples

- **The time this class starts.**
  - Official: 10.00am
  - Range: [10.00am, 10.30am]

- **The volume of water in a cup.**
  - Regular cup: 200ml
  - Range: [150ml, 250ml]

ATTENDANCE CHECK

Question: Write an example of what you think is a continuous random variable.

Probability **Density** Function

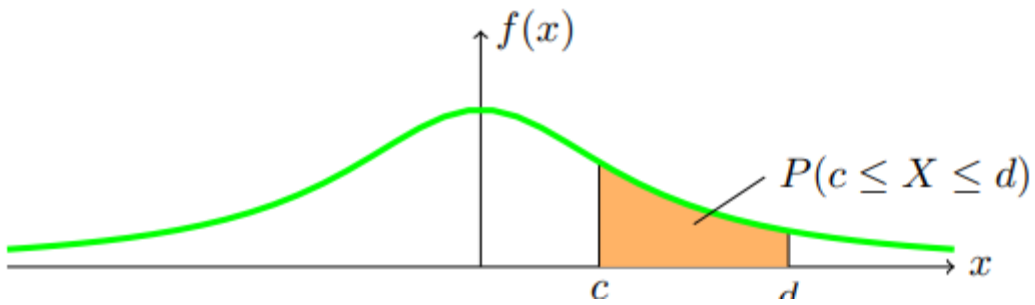
- Probability **Density** Function (p**df**)
  - Not **pmf**!

- $P(c \leq X \leq d) = \int_c^d f(x)dx.$
  - $f(x)$ is the **pdf**.

- **Properties:**
  - $f(x) \geq 0$ f(x) is non-negative
  - $\int_{-\infty}^{\infty} f(x)dx = 1$ The probability between –inf to inf is 1.

Probability Density Function (2)

$P(c \leq X \leq d)$ = area under the graph between c and d.



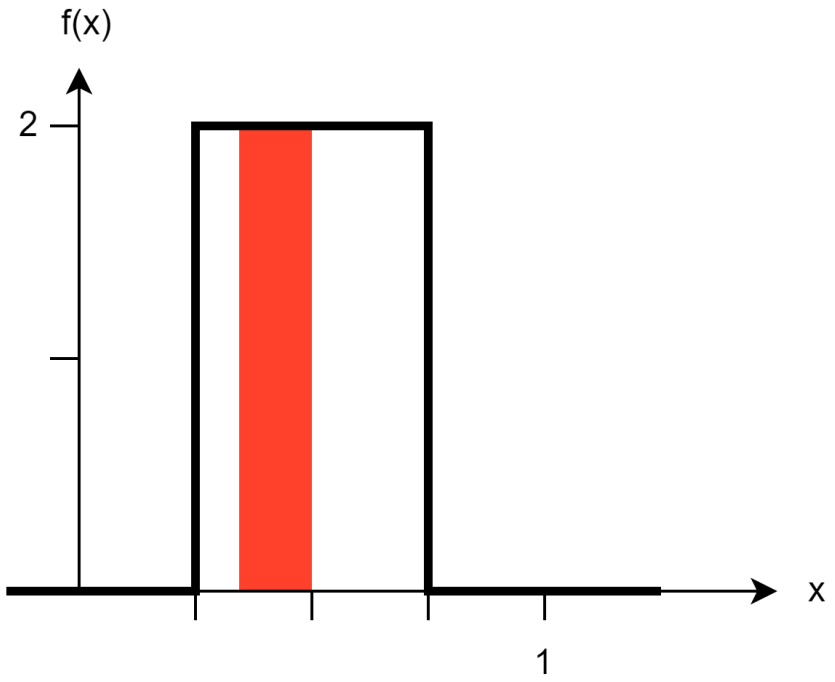
- Unlike pmf $p(x)$, **<u>pdf $f(x)$ IS NOT probability.</u>**
  - $f(x)$ <u>CAN BE</u> more than 1. Probability cannot.
  - Probability $P(c \leq X \leq d) = \int_c^d f(x)dx$.
  - $P(X = a) = 0$ because $\int_a^a f(x)dx = 0$.

Example 1

Problem. Suppose random variable X has $f(x) = 2$ on range $\left[\frac{1}{4}, \frac{3}{4}\right]$. Draw a density graph and compute $P(0.3 \leq X \leq 0.5)$.

Ans.

$$P(.3 \leq X \leq .5) = \int_{.3}^{.5} 2\, dx = 0.4.$$



Example 2

Problem. Let X be a random variable with range [0,2] and pdf $f(x) = Cx^2$. What is the value of C?

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

Ans. Since the total probability must be 1:

$$\int_0^2 f(x)dx = 1 \;\rightarrow\; \int_0^2 Cx^2dx = 1 \;\rightarrow\; \left.\frac{Cx^3}{3}\right|_0^2 = 1 \;\rightarrow\; \frac{8C}{3} = 1$$

$$\boxed{C = 3/8}$$

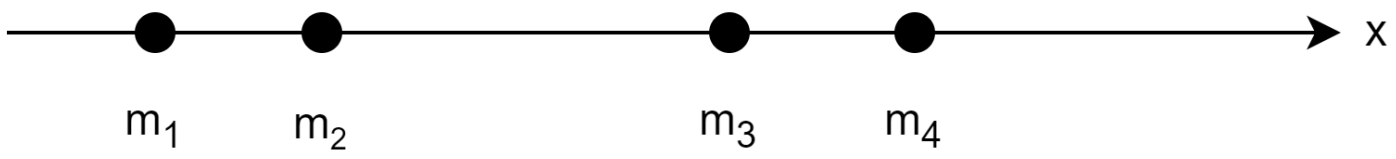
Example 3

Problem. From last slide $f(x) = \frac{3x^2}{8}$ on $[0,2]$. Find $P(X \geq 1)$.

Ans. $P(X \geq 1) = \int_1^2 f(x)dx = \int_1^2 \frac{3x^2}{8} dx = \frac{x^3}{8}\Big|_1^2 = 1 - \frac{1}{8} = \boxed{\frac{7}{8}.}$

Mass vs. Density

- Discrete: mass function



- Continuous: density

Cumulative Distribution Function

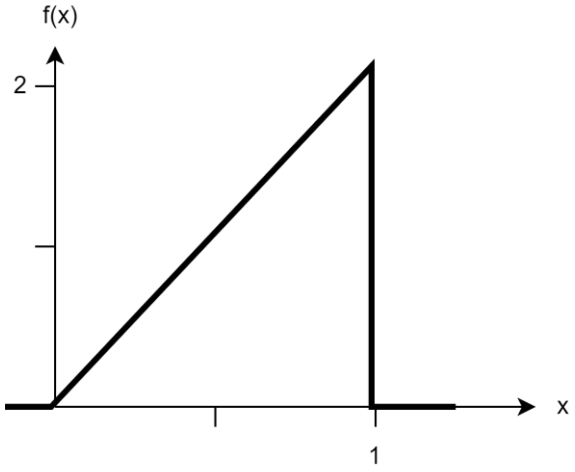
- **Similar** to the discrete random variable.
- $F(b) = P(X \leq b) = \int_{-\infty}^{b} f(x)dx$ where $f(x)$ is the pdf of X.
- $F(x)$ is commonly used for continuous random variable.

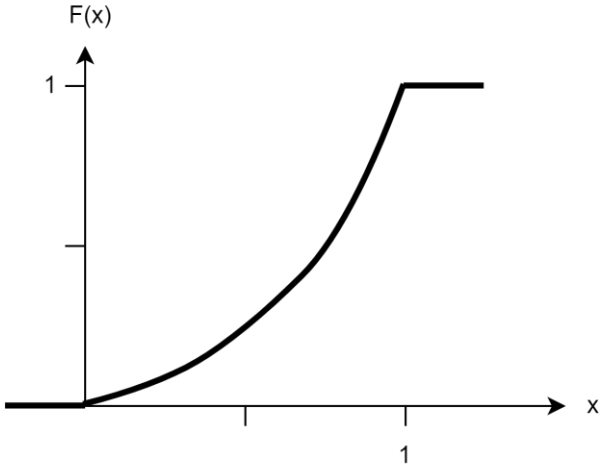
"X has distribution of F(x)"

Example 4

Problem. Suppose X is a random variable on $[0,1]$ with $f(x) = 2x$. Find CDF.

Ans. $f(x) = 2x$ on [0,1] ➔ $F(a) = \int_0^a 2x\,dx = a^2$.





Properties of CDF

1. $F(x) = P(X \leq x)$
2. $0 \leq F(x) \leq 1$
3. $F(X)$ is non-decreasing.
4. $\lim_{x \to \infty} = 1$ and $\lim_{x \to -\infty} 0$.
5. $P(a \leq X \leq b) = F(b) - F(a)$
6. $F'(x) = f(x)$.

Quiz 1

Let X be a random variable with pdf $f(x) = \begin{cases} 1, & 0 \le x \le 0.5 \\ 0.5, & 1 \le x \le 2. \\ 0, & elsewhere \end{cases}$

1. Draw pdf and cdf.

2. Calculates:
   1. $P(0 \le x \le 2)$.
   2. $P(0.25 \le x \le 1.25)$.

10 MIN BREAK

Uniform Distribution

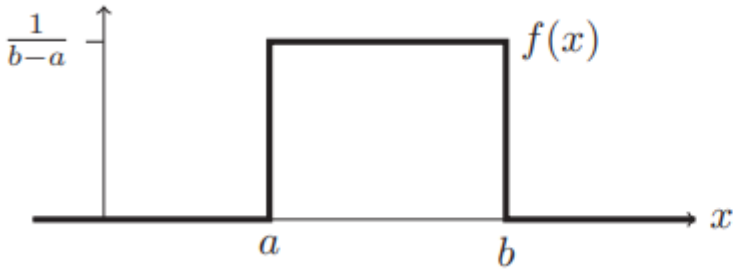
Parameters: a, b.

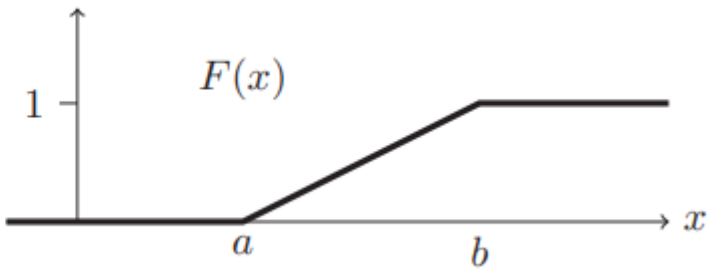
Density: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$.

Range: [a,b]

Distribution: $F(x) = \frac{x-a}{b-a}$ for $a \leq x \leq b$.

Notation: U(a,b)





Example

Model: All outcomes in the range have equal probability.

- Wheel spinner
- Measurement (rounding error).
- Etc.

Exponential Distribution

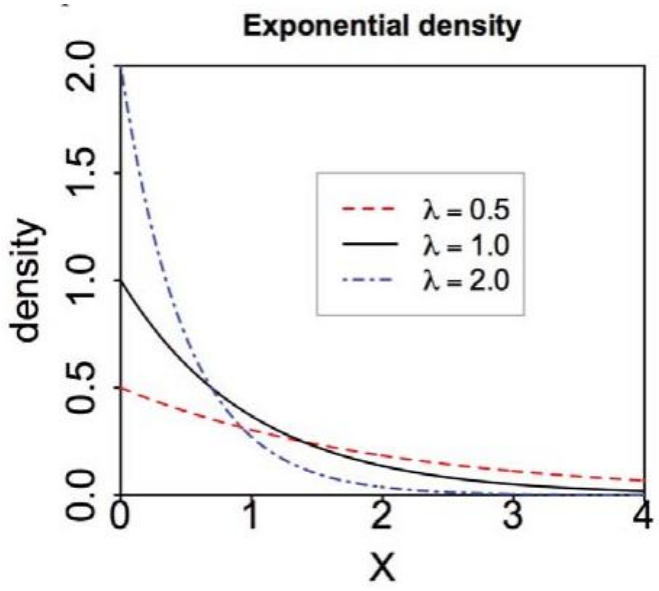
Parameters: λ.

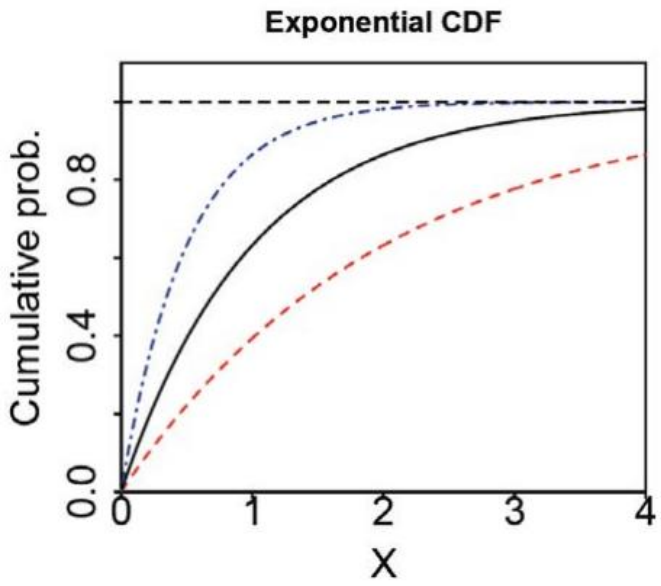
Range: $[0, \infty]$

Notation: $\exp(\lambda)$

Density: $f(x) = \lambda e^{-\lambda x}$ for $0 \leq x$.

Distribution: $F(x) = 1 - e^{-\lambda x}$ for $0 \leq x$.





CDF Integral:

$$\int \lambda e^{-\lambda x} dx = \lambda \int e^{-\lambda x}$$

Set $u = -\lambda x, du = -\lambda dx$

$$= \lambda \int e^{u} \left(\frac{1}{-\lambda}\right) du$$

$$= -\int e^{u} du = -e^{u} + C = -e^{-\lambda x} + C$$

Example

Model: Waiting time until something occurs.

1. The start time of this class (probably).
2. Waiting time for taxi (assuming no app).

Memoryless (again)

- Exponential distribution is **memoryless**.
  - That is, if you waits 5 minutes, the expected waiting time does not change.

- In maths: $P(X > s + t | X > s) = P(X > t)$.

Proof of memorylessness: Since $(X > s+t) \cap (X > s) = (X > s+t)$ we have

$$P(X > s+t \,|\, X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{\mathrm{e}^{-\lambda(s+t)}}{\mathrm{e}^{-\lambda s}} = \mathrm{e}^{-\lambda t} = P(X > t). \text{ QED}$$

Normal Distribution

Parameters: μ, σ.

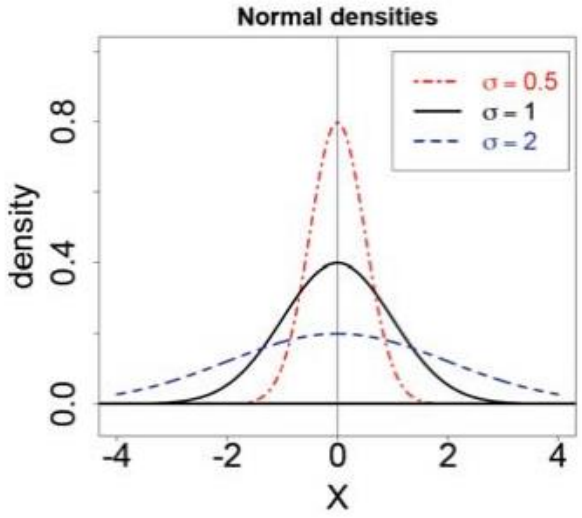
Range: $[-\infty, \infty]$

Notation: $\mathrm{N}(\mu, \sigma^2)$

Density: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$

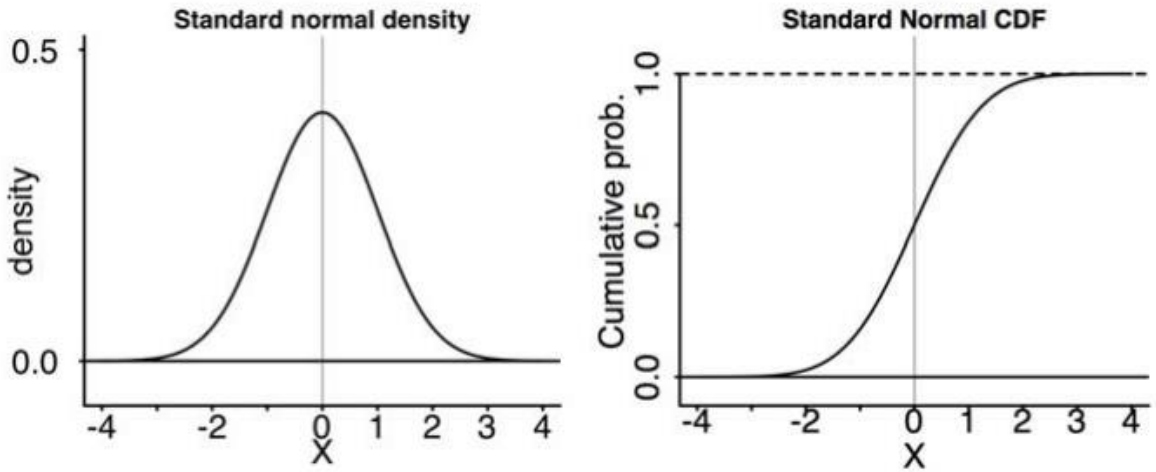
Distribution: No formula. Use lookup table.

"Bell curve"



Standard Normal Distribution

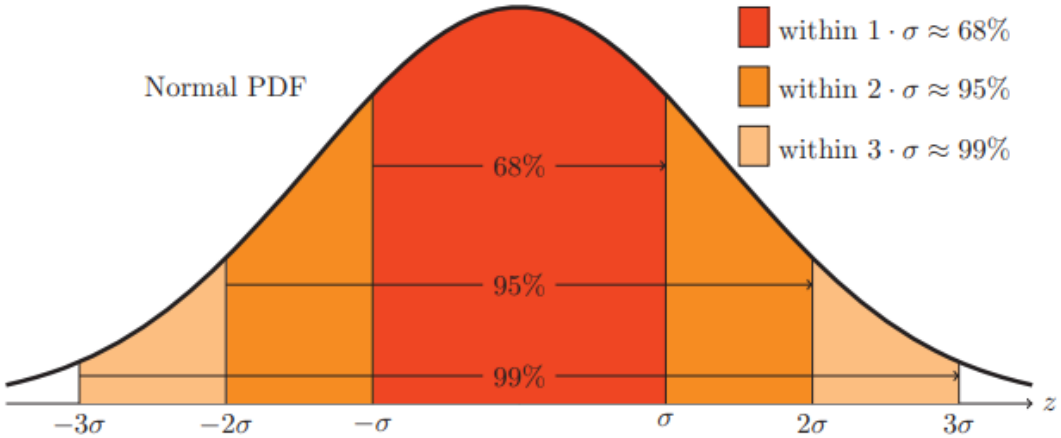
- **Standard Normal Distribution** refers to $N(0,1)$.
  - Z refers to standard normal random variable
  - $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ for standard normal density
  - $\Phi(z)$ for standard normal distribution



Normal probabilities

- Note that the PDF is symmetric.
- $P(-1 \leq Z \leq 1) \approx 0.68.$
- $P(-1 \leq Z \leq 0) = P(0 \leq Z \leq 1) \approx 0.34.$

- https://en.wikipedia.org/wiki/Standard_normal_table#Cumulative_from_minus_infinity_to_Z



Example

Model: measurement error, ability, height, **average of lots of data.**

1. The amount of water in a specific cup filled to brim.
2. **Central Limit Theory** ➔ Next Week.

Quiz 2

Using the standard normal table, calculate the following:

1. $P(Z \leq 1)$
2. $P(Z \geq 2.5)$
3. $P(-1.5 \leq Z \leq 1.5)$

z	+ 0.00	+ 0.01	+ 0.02	+ 0.03	+ 0.04	+ 0.05	+ 0.06	+ 0.07	+ 0.08	+ 0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56360	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91308	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520

LUNCH BREAK